Infrared Einstein Response from a Renormalized Spectral Entropy Functional

Jérôme Beau[1*]

[1*]Independent Researcher, France.

Corresponding author(s). E-mail(s): jerome.beau@cosmochrony.org;

Abstract

We investigate the covariant metric variation of a minimal spectral entropy functional $S_\Pi[g] = \frac{1}{2}\log\det' A_g$ defined by a Laplace-type elliptic operator on a four-dimensional Riemannian manifold. Using heat-kernel methods and zeta regularization, we analyze the renormalized functional and determine the geometric structure of the resulting effective stress tensor.

We show that the renormalized spectral variation decomposes into a local two-derivative Einstein tensor term, a cosmological term, local four-derivative quadratic curvature contributions, non-local form factors, and a conformal anomaly piece. The Einstein tensor arises from the counterterm associated with the a_2 heat-kernel pole, in direct analogy with induced gravity in the sense of Sakharov.

In the weak-curvature regime where all dimensionless curvature invariants satisfy $R\,\ell_{\rm sp}^2 \ll 1$, the local two-derivative contribution dominates over higher-derivative and non-local terms. The induced Newton coupling scales parametrically with the spectral cutoff as $G_N \sim 16\pi^2\ell_{\rm sp}^2$, up to scheme-dependent factors.

These results show that the infrared-dominant local part of the renormalized spectral entropy variation reproduces the Einstein tensor. We further show that the ultraviolet completion of this infrared sector is uniquely determined, under a coherence extensivity hypothesis, to be of determinantal Born–Infeld type: the action density takes the Eddington-inspired form $\sqrt{-\det\left(g_{\mu\nu} + \ell_{\rm sp}^2\,R_{\mu\nu}\right)} - \sqrt{-g}$. This mechanism does not rely on a specific microscopic completion, but follows from general properties of the renormalized spectral functional. In this sense the resulting Einstein response appears as the universal two-derivative infrared sector of the effective geometric dynamics, and its Born–Infeld completion as the unique admissible spectral ultraviolet extension. Geometric gravity therefore appears as an effective low-energy response of spectral operator structure rather than as a fundamental dynamical input.

Keywords: Spectral geometry, Heat-kernel expansion, Induced gravity, Zeta regularization, Effective action, Einstein tensor, Born–Infeld completion, Renormalization

Contents

1 Introduction

Gravity is traditionally described as a geometric phenomenon. In general relativity, the gravitational interaction is encoded in the curvature of spacetime, and the Newtonian potential arises as a weak-field limit of the Einstein equations. This geometric formulation is remarkably successful. It nevertheless leaves open the conceptual question of whether gravitational dynamics must be fundamental, or whether it may arise as an effective response of a more primitive operator-based structure [1].

In parallel, operator-theoretic and spectral approaches have clarified how elliptic operators encode geometric and physical information. Functional determinants, resolvents, and heat-kernel asymptotics play a central role in quantum field theory, statistical mechanics, and spectral geometry [2, 3]. Variations of logarithmic determinants generate non-local responses through the inverse operator. This suggests a bridge between entropy-like functionals and long-range interaction.

In a previous analysis [4], we investigated a minimal mechanism by which a Newtonian interaction can emerge from the variation of a projective entropy functional in three spatial dimensions. For a positive Laplace-type operator A, the functional

$$S_\Pi[A] = \frac{1}{2} \log \det{}' A \tag{1}$$

has the weak-perturbation variation

$$\delta S_\Pi = \frac{1}{2} \operatorname{Tr}\big(A^{-1} \delta A\big) , \tag{2}$$

so that the Green kernel governs the spatial structure of the response. In three dimensions, the Green kernel of a Laplace-type operator exhibits a universal $1/r$ decay. Consequently, the entropy response inherits a Newtonian long-range profile without assuming a force law.

The goal of the present paper is to develop a covariant extension of this mechanism and to identify, in a controlled infrared regime, the leading local geometric content of the *renormalized* metric variation. We do not claim an exact equality between the metric variation of the bare determinant and Einstein's equations. We also do not discard non-local contributions, higher-derivative curvature terms, or the conformal anomaly. Instead, our central claim is the following.

Central claim (renormalized IR locality).

In four dimensions, for a minimal Laplace-type operator and in the weak-curvature regime where all dimensionless curvature invariants satisfy $R\,\ell_{\rm sp}^2 \ll 1$, the infrared-dominant *local* two-derivative part of the renormalized metric variation $\delta S_\Pi^{ren}/\delta g^{\mu\nu}$ is proportional to the Einstein tensor $G_{\mu\nu}$. The associated coefficient defines an induced Newton coupling whose scaling is set by the spectral cutoff scale $\ell_{\rm sp}$. The Einstein term arises from renormalization of the geometric sector, via the counterterm associated with the a_2 heat-kernel pole, in the standard sense of induced gravity. The structural necessity of non-injectivity for any genuinely emergent description is established in [5].

The present paper is a middle link in a derivation chain. The Laplace-type operator $A_g = -\nabla_g^2$ is taken here as the starting point. Its identification with the Cosmochrony admissibility operator L_Π is established by the chain: Q5a [6] (Mosco convergence $\mathcal{E}_q \to L_\Pi$), Q5b [7] (BFS stratification lifts L_Π to L_{eff} on $\mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$), and Q9 [8] ([H-lift] proved unconditionally). The identification $L_\Pi = A_g$ is therefore structural, not a working hypothesis. The present paper establishes the infrared consequence: any such operator yields, after renormalization, a dominant Einstein response.

Operationally, this result identifies a robust hierarchy. Local four-derivative contributions (quadratic curvature variations) and non-local form factors are suppressed in the infrared by additional powers of ℓ_{sp}^2, while the conformal anomaly remains as a distinct ultraviolet-sensitive contribution whose presence is tracked explicitly.

The remainder of the paper is organized as follows. In Section 2 , we define the covariant projective entropy functional and set up the minimal Laplace-type operator choice. In Section 3 , we summarize the heat-kernel expansion and the role of Seeley–DeWitt coefficients in the pole structure of zeta regularization. In Section 4 , we define the renormalized spectral stress tensor and decompose it by derivative order into Einstein, quadratic-curvature, non-local, and anomaly pieces. In Section 4.5 , we formulate the infrared hierarchy as a covariant derivative expansion and state the dominance criterion $R\,\ell_{\text{sp}}^2 \ll 1$. In Section 5 , we discuss the induced-gravity interpretation and the identification of the effective Newton coupling. We conclude in Section 7 with limitations and outlook.

2 Covariant projective entropy functional

2.1 Ellipticity and minimal Laplace-type operator

Let (M, g) be a smooth four-dimensional Riemannian manifold. We consider a second-order differential operator acting on scalar functions.

The analysis relies crucially on ellipticity. For a differential operator of order two, ellipticity means that its principal symbol is invertible away from the zero section.

We choose the minimal Laplace-type operator

$$A_g = -\nabla_g^2, \tag{3}$$

where ∇_g is the Levi-Civita connection. Its principal symbol is

$$\sigma_2(A_g)(x, k) = g^{\mu\nu}(x) k_\mu k_\nu. \tag{4}$$

On a Riemannian manifold, $g^{\mu\nu} k_\mu k_\nu > 0$ for all nonzero k. Therefore A_g is elliptic.

The principal symbol fully determines the local propagation structure. In particular, the metric enters exclusively through $\sigma_2(A_g)$. Lower-order terms do not affect ellipticity.

Ellipticity ensures:

- existence of the resolvent $(A_g + \lambda)^{-1}$ for $\lambda > 0$,
- discreteness of the spectrum on compact manifolds,
- local regularity of the Green kernel,
- validity of the short-time heat-kernel expansion,

- meromorphic continuation of the spectral zeta function.

More general scalar Laplace-type operators may include a curvature coupling,

$$A_g(\xi) = -\nabla_g^2 + \xi R. \tag{5}$$

The additional term ξR modifies only the zeroth-order endomorphism part of the operator and does not alter its principal symbol. It therefore does not affect ellipticity.

In this work we set $\xi = 0$. This is the minimal operator choice in which the effective geometry is entirely encoded in the principal symbol, without additional curvature input introduced at lower order.

The conformal value $\xi = 1/6$ cancels the a_2 coefficient in four dimensions and suppresses the induced Einstein–Hilbert counterterm. The present analysis focuses on the minimal elliptic case $\xi = 0$, for which the a_2 contribution is nonvanishing.

2.2 Regularized determinant and renormalization setup

We define the projective entropy functional by

$$S_\Pi[g] \equiv \frac{1}{2} \log \det{}' A_g, \tag{6}$$

where the prime denotes omission of zero modes, if any.

The determinant is understood in the zeta-regularized sense,

$$\log \det{}' A_g = -\zeta'_{A_g}(0), \tag{7}$$

with

$$\zeta_{A_g}(s) = \frac{1}{\Gamma(s)} \int_0^\infty t^{\,s-1}\, \mathrm{Tr}\big(e^{-tA_g}\big)\; dt, \tag{8}$$

initially defined for $\mathrm{Re}(s)$ sufficiently large and continued meromorphically to $s = 0$.

In four dimensions, the small-t expansion of the heat trace contains ultraviolet divergences. Renormalization is implemented by subtracting the pole terms associated with the heat-kernel coefficients and introducing a renormalization scale μ.

The resulting renormalized functional $S_\Pi^{ren}[g;\mu]$ is finite but contains scheme-dependent local geometric counterterms. These counterterms are crucial for the emergence of the Einstein–Hilbert structure discussed below.

2.3 First variation and resolvent trace

For a perturbation $A_g \to A_g + \delta A_g$, the first variation of the determinant satisfies formally

$$\delta S_\Pi = \frac{1}{2}\, \mathrm{Tr}\big(A_g^{-1}\, \delta A_g\big)\,. \tag{9}$$

This identity shows that the response of the functional is governed by the resolvent kernel. In three spatial dimensions, this mechanism produces a universal $1/r$ Green profile, leading to a Newtonian response without assuming a fundamental force law.

In the present covariant four-dimensional setting, the metric variation defines a renormalized spectral stress tensor,

$$\mathcal{T}_{\mu\nu}^{\Pi,ren} \equiv -\frac{2}{\sqrt{g}}\frac{\delta S_{\Pi}^{\text{ren}}}{\delta g^{\mu\nu}}. \tag{10}$$

Its precise geometric decomposition depends on the heat-kernel structure of A_g. In particular, local two-derivative, local four-derivative, non-local, and anomaly contributions will appear with distinct origins. The classification of these terms and their infrared hierarchy is developed in the following sections.

3 Heat-kernel expansion and Seeley–DeWitt structure

3.1 Short-time heat-kernel expansion

For an elliptic Laplace-type operator A_g on a smooth four-dimensional Riemannian manifold, the heat trace admits a short-time asymptotic expansion as $t \to 0^+$ of the form

$$\mathrm{Tr}\big(e^{-tA_g}\big) \sim \frac{1}{(4\pi t)^2}\sum_{k\geq 0} t^k \int_M d^4x\,\sqrt{g}\,a_{2k}(x). \tag{11}$$

The coefficients $a_{2k}(x)$ are local curvature invariants constructed from the metric and its derivatives. They are the Seeley–DeWitt coefficients of the operator [2, 9, 10].

For the minimal scalar Laplacian ($\xi = 0$), with vanishing endomorphism $E = 0$ and bundle curvature $\Omega = 0$, the first coefficients are

$$a_0(x) = 1, \tag{12}$$

$$a_2(x) = \frac{1}{6}R(x), \tag{13}$$

$$a_4(x) = \frac{1}{(4\pi)^2}\frac{1}{360}\left(\frac{5}{3}R^2 - R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right). \tag{14}$$

Thus, in the decomposition

$$a_4 = \alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}, \tag{15}$$

the explicit numerical coefficients are

$$\alpha_1 = \frac{5}{3\cdot 360(4\pi)^2}, \quad \alpha_2 = -\frac{1}{360(4\pi)^2}, \quad \alpha_3 = \frac{1}{360(4\pi)^2}. \tag{16}$$

In four dimensions, the quadratic curvature invariants are not all independent. The Gauss–Bonnet density

$$\mathcal{G} = R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - 4R_{\mu\nu}R^{\mu\nu} + R^2 \tag{17}$$

is topological and does not contribute to the local equations of motion. The chosen basis $\{R^2, R_{\mu\nu}R^{\mu\nu}, R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\}$ is therefore equivalent, up to a Gauss–Bonnet term, to the commonly used basis $\{R^2, R_{\mu\nu}R^{\mu\nu}, \mathcal{G}\}$.

3.2 Pole structure in four dimensions

Substituting the heat expansion into the definition of $\zeta_{A_g}(s)$ shows that in $d = 4$ the coefficients a_0 and a_2 generate simple poles at $s = 2$ and $s = 1$, respectively.

More precisely:

- a_0 controls the cosmological-constant counterterm,
- a_2 controls the Einstein–Hilbert counterterm,
- a_4 controls quadratic curvature counterterms.

After subtraction of the pole terms and introduction of the renormalization scale μ, the renormalized functional contains local geometric counterterms whose coefficients depend on μ.

3.3 Origin of the Einstein tensor

The crucial structural point is the following.

The coefficient $a_2(x) = R/6$ implies that the pole subtraction generates a local term proportional to

$$\int d^4x \sqrt{g}\, R. \tag{18}$$

Upon metric variation, this counterterm produces the Einstein tensor,

$$\frac{\delta}{\delta g^{\mu\nu}} \int d^4x \sqrt{g}\, R \;\propto\; G_{\mu\nu}. \tag{19}$$

Therefore, the local two-derivative Einstein term in the renormalized spectral stress tensor arises from the counterterm associated with the a_2 heat-kernel pole.

It does not arise from the bare finite part of $\zeta'_{A_g}(0)$ alone. Rather, it is generated through renormalization of the geometric sector, in the standard sense of induced gravity.

This mechanism is structurally identical to Sakharov's induced-gravity construction [11], with the spectral cutoff scale playing the role of the ultraviolet regulator.

Higher-derivative local contributions originate from a_4 and yield four-derivative tensors (Bach-like structures) upon variation. Non-local form factors arise from the finite part of the determinant and will be treated separately.

4 Metric variation and renormalized spectral stress tensor

4.1 Definition of the renormalized spectral stress tensor

The renormalized projective entropy functional $S_{\Pi}^{\mathrm{ren}}[g;\mu]$ defines a covariant stress tensor by

$$\mathcal{T}_{\mu\nu}^{\Pi,\mathrm{ren}} \equiv -\frac{2}{\sqrt{g}}\frac{\delta S_{\Pi}^{\mathrm{ren}}}{\delta g^{\mu\nu}}. \tag{20}$$

Formally, prior to renormalization, the first variation reads

$$\delta S_{\Pi} = \frac{1}{2}\,\mathrm{Tr}\big(A_g^{-1}\delta A_g\big)\,, \tag{21}$$

so that the response is governed by the resolvent kernel. After renormalization, local counterterms contribute additional geometric variations.

In four dimensions, and for the minimal Laplacian $\xi = 0$, the metric variation admits the general decomposition

$$\frac{\delta S_{\Pi}^{\mathrm{ren}}}{\delta g^{\mu\nu}} = c_{\mathrm{E}H}(\mu)\,G_{\mu\nu} + c_{\Lambda}(\mu)\,g_{\mu\nu} + \beta(\mu)\,B_{\mu\nu} + T_{\mu\nu}^{\mathrm{non}\text{-}loc} + A_{\mu\nu}. \tag{22}$$

Each term has a distinct geometric origin.

4.2 Local two-derivative sector

The Einstein tensor contribution arises from the renormalized Einstein–Hilbert counterterm generated by the a_2 pole. From Section 3, the subtraction of the a_2 divergence produces a local term proportional to

$$\int d^4x\,\sqrt{g}\,R. \tag{23}$$

Its metric variation yields

$$\frac{\delta}{\delta g^{\mu\nu}}\int d^4x\,\sqrt{g}\,R = -\sqrt{g}\,G_{\mu\nu}. \tag{24}$$

Therefore,

$$c_{\mathrm{E}H}(\mu) = \frac{1}{16\pi G_N(\mu)}, \tag{25}$$

where $G_N(\mu)$ is the induced Newton coupling.

The cosmological constant term originates from the a_0 pole and is proportional to $\int\sqrt{g}$.

4.3 Local four-derivative sector

The coefficient a_4 generates quadratic curvature counterterms of the form

$$\int d^4x\,\sqrt{g}\,\left(\alpha_1 R^2 + \alpha_2 R_{\mu\nu}R^{\mu\nu} + \alpha_3 R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma}\right). \tag{26}$$

Their variation produces fourth-order differential tensors, which we collectively denote by $B_{\mu\nu}$. These are Bach-like structures containing four derivatives of the metric.

4.4 Non-local form factors and anomaly

The finite part of the determinant generates non-local contributions. In covariant form, these can be expressed through curvature-dependent form factors of the type

$$R\,\mathcal{F}\left(\frac{\Box}{\mu^2}\right)R, \tag{27}$$

and similar structures, as developed by Barvinsky and Vilkovisky [12, 13]. Their metric variation yields non-local tensors $T_{\mu\nu}^{\text{non-loc}}$.

In addition, renormalization induces a conformal trace anomaly. For a minimally coupled scalar field in four dimensions, the trace of the anomaly satisfies [2, 14]

$$g^{\mu\nu}A_{\mu\nu} = \frac{1}{(4\pi)^2}\frac{1}{360}\left(R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} - R_{\mu\nu}R^{\mu\nu} + \frac{1}{3}\Box R\right). \tag{28}$$

The local tensor structure of $A_{\mu\nu}$ depends on the renormalization scheme, but its trace is fixed by this universal expression. The anomaly contributes at four-derivative order and remains subleading in the weak-curvature infrared regime $R\,\ell_\chi^2 \ll 1$.

4.5 Infrared hierarchy

Consider the regime in which all dimensionless curvature invariants satisfy

$$R\,\ell_{\text{sp}}^2 \ll 1, \qquad R_{\mu\nu}\,\ell_{\text{sp}}^2 \ll 1. \tag{29}$$

In this regime, the derivative expansion is controlled by the small parameter $R\,\ell_{\text{sp}}^2$. Since $B_{\mu\nu}$ is quadratic in curvature, its contribution scales as

$$\frac{\beta\,B_{\mu\nu}}{c_{\text{EH}}\,G_{\mu\nu}} \sim R\,\ell_{\text{sp}}^2. \tag{30}$$

Therefore, for $R\,\ell_{\text{sp}}^2 \ll 1$, the local two-derivative Einstein term dominates over the four-derivative sector. Non-local form factors reduce to suppressed derivative corrections.

This establishes the infrared dominance of the Einstein tensor within the renormalized spectral stress tensor.

5 Identification of Newton's constant and spectral scaling

5.1 Running of the Einstein–Hilbert coefficient

In four dimensions, the subtraction of the a_2 pole generates a renormalization-scale dependence in the Einstein–Hilbert coefficient.

For the minimal scalar Laplacian with $a_2 = R/6$, the scale dependence of the induced coupling satisfies

$$\frac{d}{d \log \mu} \left(\frac{1}{16\pi G_N(\mu)} \right) = \frac{1}{(4\pi)^2} \frac{1}{6}. \tag{31}$$

This relation follows directly from the residue of the a_2 pole in the zeta-regularized determinant. It expresses the standard induced-gravity running generated by a minimal scalar degree of freedom [3, 14].

The absolute normalization of G_N depends on the ultraviolet completion or cutoff prescription. However, its parametric dependence is fixed by dimensional analysis.

5.2 Spectral cutoff and induced scale

Let Λ_{UV} denote the ultraviolet cutoff scale associated with the spectral regularization. Identifying Λ_{UV} with the inverse spectral resolution scale,

$$\Lambda_{UV} \sim \ell_{\mathrm{sp}}^{-1}, \tag{32}$$

the induced Einstein–Hilbert coefficient scales as

$$\frac{1}{16\pi G_N} \sim \frac{\Lambda_{UV}^2}{16\pi^2}. \tag{33}$$

Therefore,

$$G_N \sim 16\pi^2 \, \ell_{\mathrm{sp}}^2, \tag{34}$$

up to scheme-dependent numerical factors and possible multiplicities from additional degrees of freedom.

This scaling relation provides the bridge between the spectral resolution scale ℓ_{sp} and the effective gravitational coupling.

5.3 Infrared consistency

The infrared hierarchy established in Section 4.5 requires

$$R \, \ell_{\mathrm{sp}}^2 \ll 1. \tag{35}$$

Using the scaling above, this condition is equivalent to

$$R \ll G_N^{-1}, \tag{36}$$

which is precisely the weak-curvature regime in which classical general relativity is valid.

Thus the dominance of the Einstein tensor in the renormalized spectral stress tensor is self-consistent in the classical infrared domain.

6 Ultraviolet completion: uniqueness of the Born–Infeld structure

The infrared hierarchy of Section 4.5 ensures that the Einstein response dominates for $R\,\ell_{\rm sp}^2 \ll 1$. In the ultraviolet regime $R\,\ell_{\rm sp}^2 \sim 1$, higher-derivative corrections become comparable to the Einstein–Hilbert term and the expansion breaks down. The question is whether the full non-perturbative effective action is determined by the same spectral structure, or requires independent input.

In the scalar Born–Infeld theory derived in [15], the saturation bound $|\partial_t\chi| \le c_{\rm BI}$ uniquely selects a square-root action density. We now show that the tensorial extension of this result to the gravitational sector is similarly rigid: under admissibility conditions inherited from the framework of [5, 16], the determinantal Born–Infeld form is the *unique* admissible spectral tensorial completion of the Einstein–Hilbert sector.

6.1 Setup and admissibility conditions

Define the mixed Ricci endomorphism

$$X^{\mu}{}_{\nu} \;=\; \ell_{\rm sp}^2\, g^{\mu\alpha}\, R_{\alpha\nu}\,. \tag{37}$$

A local covariant completer takes the form

$$\mathcal{C}(g,R) \;=\; \sqrt{-g}\,\big[\,\Phi(X) - 1\,\big]\,, \tag{38}$$

where Φ is a smooth scalar function of the endomorphism X. The effective gravitational action is then

$$S_{\rm coh}^{\rm full}[g] \;=\; \frac{1}{16\pi G_N}\int \mathcal{C}(g,R)\,\mathrm{d}^4x\,,$$

with $G_N \sim 16\pi^2\ell_{\rm sp}^2$ as established in Section 5.

The following conditions encode the structural constraints that any admissible tensorial completion must satisfy.

- **(C1)** *Spectral covariance.* $\Phi(X)$ depends on X only through its eigenvalues: $\Phi(UXU^{-1}) = \Phi(X)$ for all $U \in \mathrm{O}(d)$.
- **(C2)** *Einstein infrared limit.* For weak curvature ($\|X\| \ll 1$): $\Phi(I + \varepsilon X) = 1 + \frac{1}{2}\,\varepsilon\,\mathrm{tr}\,X + O(\varepsilon^2)$, reproducing the Einstein–Hilbert density at leading order.
- **(C3)** *Algebraic Born–Infeld saturation.* $\Phi(X) \to 0$ as any eigenvalue of $(I+X)$ approaches zero. The completer density vanishes at maximal admissible curvature rather than diverging, in accordance with the scalar saturation bound of [15].
- **(C4)** *Spectral isotropy.* Φ treats all eigenvalue directions symmetrically: $\Phi(X) = \varphi(\lambda_1,\ldots,\lambda_d)$ with φ symmetric.
- **(C5)** *Coherence extensivity.* Define $S_{\rm coh} = \log\Phi$. For admissibly independent sectors X_1, X_2 (in the sense of Axiom A3 of [16]): $S_{\rm coh}(X_1 \oplus X_2) = S_{\rm coh}(X_1) + S_{\rm coh}(X_2)$.

Conditions (C1)–(C4) are direct consequences of the covariance, infrared, and boundedness properties established in the spectral framework of Sections 2–5 and in [15]. Condition (C5) is the only additional input; its derivation from the non-factorisability axiom A3 of [16] is the subject of the following hypothesis.

Hypothesis 1 (**H-ext**: admissible coherence extensivity)**.** *Let X_1 and X_2 be curvature endomorphisms associated with admissibly independent sectors in the sense of Axiom A3 of [16]: no cross-admissibility constraint couples the two sectors after projection locking. Then the coherence functional is extensive:*

$$S_{\mathrm{coh}}(X_1 \oplus X_2) = S_{\mathrm{coh}}(X_1) + S_{\mathrm{coh}}(X_2)\,. \tag{39}$$

Remark. Axiom A3 of [16] forbids premature factorisation of the admissible fibre prior to projection locking. After locking, sectors that carry no mutual admissibility constraint contribute independently to the total coherence cost. The extensivity of S_{coh} is the post-locking counterpart of pre-locking non-factorisability: precisely because A3 prevents premature selection, the residual cost after locking decomposes additively over independently resolved sectors. This is structurally analogous to the extensivity of thermodynamic entropy for non-interacting subsystems. An analytical derivation of [H-ext] from A3 would promote the uniqueness theorem below to an unconditional result; pending such a derivation, [H-ext] occupies a status analogous to [H-color] in the spectral admissibility sub-programme [17].

6.2 Uniqueness of the tensorial completion

The uniqueness argument proceeds in three lemmas. Lemmas 2 and 3 are unconditional; Lemma 1 is conditional on [H-ext] and carries the only conditionality in the chain.

Lemma 1 (Spectral decomposition from coherence extensivity)**.** *Assume* [H-ext]*. Then conditions* (C1)*,* (C4)*, and* (C5) *together force*

$$S_{\mathrm{coh}}(X) = \sum_{i=1}^{d} h(\lambda_i)\,, \tag{40}$$

for a single smooth function $h : (-1, +\infty) \to \mathbb{R}$, where $\lambda_1, \ldots, \lambda_d$ are the eigenvalues of X.

Proof The extensivity of S_{coh} implies $\Phi(X_1 \oplus X_2) = \Phi(X_1)\,\Phi(X_2)$ by exponentiation. Write X in diagonal form $X = \mathrm{diag}(\lambda_1, \ldots, \lambda_d)$ relative to the g-orthonormal eigenbasis. Apply the multiplicativity of Φ iteratively to the rank-1 decomposition $X = \lambda_1 P_1 \oplus \cdots \oplus \lambda_d P_d$ (where P_i are rank-1 projectors onto the eigenspaces): $\Phi(X) = \prod_i \Phi(\lambda_i P_i)$. Taking logarithms gives (40) with $h(\lambda) = \log \Phi(\lambda P)$ for any unit projector P. Condition (C4) ensures that h does not depend on the choice of eigendirection. □

Lemma 2 (One-dimensional reduction to scalar BI)**.** *Under conditions* (C2) *and* (C3)*, the sector function h in* (40) *is uniquely determined:*

$$h(\lambda) = \tfrac{1}{2}\,\log(1+\lambda)\,. \tag{41}$$

Proof Infrared constraint. Condition (C2) requires, at first order: $\Phi(I+\varepsilon X) = \exp\bigl(\sum_i h(\varepsilon\lambda_i)\bigr) \approx 1+\varepsilon\, h'(0)\,\mathrm{tr}\, X$. Matching with $1+\frac{1}{2}\,\varepsilon\,\mathrm{tr}\,X$ gives $h'(0)=1/2$. Since $\Phi(0)=1$ at flat space, $h(0)=0$.

Saturation constraint. Condition (C3) requires $\Phi(X)\to 0$ as $\lambda\to -1^+$, hence $h(\lambda)\to-\infty$ as $\lambda\to -1^+$.

Reduction to the scalar sector. In the scalar Born–Infeld theory of [15], the one-dimensional completer is uniquely fixed by the saturation bound: the action density takes the form $\sqrt{1+\lambda}-1$ for a single mode of normalised field gradient λ. The associated coherence density is $\log\sqrt{1+\lambda} = \frac{1}{2}\log(1+\lambda)$, satisfying $h(0)=0$, $h'(0)=1/2$, and $h\to-\infty$ as $\lambda\to-1^+$.

Uniqueness. Any alternative $\tilde{h}$ satisfying the above can be written as $\tilde{h}(\lambda)=\frac{1}{2}\log(1+\lambda)+ r(\lambda)$ with $r(0)=0$, $r'(0)=0$, and r bounded below near $\lambda=-1$. In the one-dimensional sector, $\exp\bigl(\tilde{h}\bigr)$ must reproduce the scalar BI density $\sqrt{1+\lambda}$, whence $\exp(r(\lambda))=1$ identically and $r\equiv 0$. □

Lemma 3 (Spectral carrier identification)**.** *Among local covariant two-derivative symmetric tensors on (M,g), the mixed Ricci endomorphism $X^{\mu}{}_{\nu}=\ell_{\mathrm{sp}}^2\, g^{\mu\alpha}R_{\alpha\nu}$ is the unique admissible spectral carrier for Φ with non-degenerate eigenvalue spectrum.*

Proof Classification. At two-derivative order, the only symmetric $(0,2)$ tensors constructed covariantly from the metric and the Riemann tensor are $R_{\mu\nu}$ and $R\,g_{\mu\nu}$, together with linear combinations. Condition (C1) requires the argument of Φ to be an endomorphism $X: T_pM\to T_pM$, obtained by raising one index with the metric.

Exclusion of $f(R)\,\delta^{\mu}{}_{\nu}$. The endomorphism $X=f(R)\,\delta^{\mu}{}_{\nu}$ has a single repeated eigenvalue. Two manifolds with equal scalar curvature but distinct Ricci eigenvalue distributions would produce identical Φ, violating the spectral resolution required by admissibility. The resulting completer $\Phi(X)=\exp\bigl(d\cdot h(f(R))\bigr)$ is a function of R alone, degenerate along all eigenvalue directions.

Selection. The remaining candidate is the mixed Ricci tensor $g^{\mu\alpha}R_{\alpha\nu}$, which carries the full directional information of the Ricci curvature through its d independent eigenvalues. It is the unique minimal two-derivative endomorphism with non-trivial spectral content.

Trace modifications. The alternative $\mathcal{R}_{\mu\nu}=R_{\mu\nu}-\alpha\, g_{\mu\nu}R$ (for constant α) yields eigenvalues shifted by $-\alpha R$ relative to the Ricci eigenvalues. Such a shift modifies Φ only at subleading IR order and is absorbed into a redefinition of ℓ_{sp}^2 provided $\alpha\neq 1/d$. The identification $\mathcal{R}_{\mu\nu}=R_{\mu\nu}$ ($\alpha=0$) is the minimal choice. The residual ambiguity in α affects only higher-order corrections and is resolved by matching to the a_2 coefficient computed in Section 3. □

Theorem 1 (Uniqueness of the admissible tensorial completion)**.** *Let $\mathcal{C}(g,R)$ be a local covariant scalar density of the form* (38)*, satisfying conditions* (C1)–(C4) *and Hypothesis* [H-ext]*. Then, up to the trace ambiguity resolved by the heat-kernel analysis* *of Section 3:*

$$\mathcal{C}(g,R) \;=\; \sqrt{-g}\,\Big[\,\sqrt{\det\!\big(\delta^\mu{}_\nu + \ell_{\rm sp}^2\, g^{\mu\alpha}R_{\alpha\nu}\big)}\;-\;1\Big]\,. \tag{42}$$

Proof By Lemma 3, the spectral carrier is $X = \ell_{\rm sp}^2\, g^{-1}R$ with eigenvalues λ_i. By Lemma 1 (conditional on [H-ext]), $\Phi(X) = \exp\!\big(\sum_i h(\lambda_i)\big)$. By Lemma 2, $h(\lambda) = \frac{1}{2}\log(1+\lambda)$. Therefore:

$$\Phi(X) = \exp\!\Big(\,\tfrac{1}{2}\sum_i \log(1+\lambda_i)\Big) = \sqrt{\prod_i (1+\lambda_i)} = \sqrt{\det(I+X)}\,.$$

Substituting into (38) gives (42). □

Using $\det(g\cdot A) = \det g\cdot \det A$, the result takes the equivalent Eddington–Born–Infeld form:

Corollary 2 (Eddington–Born–Infeld action)**.** *Under the hypotheses of Theorem 1, the gravitational action takes the form*

$$S^{\rm full}_{\rm coh}[g] \;=\; \frac{1}{16\pi G_N}\int_M \Big[\,\sqrt{-\det\!\big(g_{\mu\nu} + \ell_{\rm sp}^2\, R_{\mu\nu}\big)}\;-\;\sqrt{-g}\,\Big]\,\mathrm{d}^4x\,. \tag{43}$$

6.3 Structural interpretation

Remark. The action (43) coincides in form with Eddington-inspired Born–Infeld gravitational actions studied independently in the modified gravity literature. The present derivation differs in two respects. First, the EiBI form is not postulated as a modified gravity model but derived as a structural rigidity from admissibility conditions. Second, the spectral capacity length $\ell_{\rm sp}$ is not a free parameter: it is determined by the same spectral cutoff that fixes the induced Newton constant in Section 5.

Remark. The framework involves two spectrally defined objects that must be carefully distinguished. At the *global* level, the projective entropy functional $S_\Pi[g] = \frac{1}{2}\log\det' A_g$ of Section 2 encodes the full eigenvalue spectrum of the effective Laplacian A_g and produces the infrared Einstein response (Section 4). At the *local* level, the coherence density $S_{\rm coh}(X) = \frac{1}{2}\log\det(I+X)$ encodes the eigenvalue spectrum of the mixed Ricci endomorphism and governs the ultraviolet saturation regime. Both share the algebraic form $\frac{1}{2}\log\det(\cdot)$ because both originate from the extensivity of admissible coherence: the former over eigenvalue modes of A_g, the latter over eigenvalue directions of $g^{-1}R$. Admissible spectral extensivity is therefore the common principle underlying both the infrared gravitational response and the ultraviolet Born–Infeld completion.

Remark. Lemma 2 is unconditional: it follows from the scalar Born–Infeld uniqueness of [15]. Lemma 3 is unconditional: it is a classification result in Riemannian geometry. Lemma 1 is conditional on [H-ext], whose derivation from Axiom A3 of [16] is structurally motivated but not yet analytically closed. Theorem 1 is therefore conditional on [H-ext] alone. The conditionality is sharply localised: it resides entirely in the passage from non-factorisability (Axiom A3) to extensivity of $S_{\rm coh}$.

7 Discussion and outlook

7.1 Summary of the structural result

We have constructed a covariant projective entropy functional

$$S_\Pi[g] = \frac{1}{2} \log \det' A_g, \tag{44}$$

for a minimal elliptic Laplace-type operator.

Using heat-kernel methods and zeta regularization in four dimensions, we showed that the renormalized metric variation admits the decomposition

$$\frac{\delta S_\Pi^{ren}}{\delta g^{\mu\nu}} = c_{\mathrm{E}H} G_{\mu\nu} + c_\Lambda g_{\mu\nu} + \beta B_{\mu\nu} + T_{\mu\nu}^{\text{non-}loc} + A_{\mu\nu}. \tag{45}$$

The Einstein tensor arises from the counterterm associated with the a_2 heat-kernel pole. Local four-derivative terms originate from a_4. Non-local form factors and anomaly contributions are retained explicitly.

In the weak-curvature regime $R\,\ell_{\mathrm{sp}}^2 \ll 1$, the local two-derivative Einstein term dominates over higher-derivative contributions. The induced Newton coupling scales parametrically as

$$G_N \sim 16\pi^2 \ell_{\mathrm{sp}}^2, \tag{46}$$

up to scheme-dependent factors.

The result is therefore not an exact derivation of Einstein's equations from the bare determinant. It is a structural statement about the infrared-dominant local two-derivative part of the renormalized spectral stress tensor.

7.2 Structural necessity of non-injective projection

The spectral derivation developed above shows that the renormalized variation of the projective entropy functional produces an infrared-dominant Einstein response. It is useful to clarify under which structural conditions such an emergent gravitational sector can arise.

Consider a physical description obtained through a projection

$$\Pi : F \to O,$$

mapping a space of admissible configurations F to a space of observable states O. The space F is not an external substrate: it is induced by the admissibility structure of the observable states, with its internal organisation constrained by the Born–Infeld capacity bound [5, 16]. Such projections need not be injective: distinct admissible configurations may yield the same observable outcome.

A basic structural observation is that admissible injective projections possess trivial holonomy. If the projection is injective, transport along closed loops in the projection fiber is necessarily trivial. The associated fiber connection therefore has vanishing curvature,

$$\Omega_{\mu\nu} = 0.$$

In this case the projection cannot generate additional geometric structure in the effective description. In particular, no contribution to the effective spacetime curvature can arise from projection-induced fiber curvature.

When the projection is non-injective, by contrast, the projection fiber may carry non-trivial curvature. The effective geometric response may then contain terms schematically of the form

$$R^{\text{eff}}_{\mu\nu} \supset \operatorname{tr}_{g_\Pi}(\Omega_{\mu\alpha}\Omega_\nu{}^\alpha)\,,$$

showing that curvature in the projection fiber can act as a source for effective spacetime curvature.

This observation provides a structural interpretation of emergent gravitation. If the effective description were injective, the projection would introduce no additional curvature and no gravitational sector would arise from projection structure. The existence of an emergent gravitational response therefore requires a non-injective projection [5].

Remark (Heisenberg torsion as the structural source). The non-commutativity underlying the gravitational response has a concrete realisation in the Cosmochrony spectral programme. In $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$, the generating set $S_q = \{X^{\pm1}, Y^{\pm1}\}$ does not include the central generator Z, yet $[X, Y] = Z$. The central direction is therefore not a degree of freedom of the BFS exploration but the structural obstruction to integrability of the horizontal distribution — the torsion of the sub-Riemannian geometry of Heis_3. The admissibility form $\mathcal{E}_q$, built on the perturbation of admissible states by $\rho_q(s)$, is directly sensitive to this non-commutativity. In the large-q Mosco limit [6], this torsion translates into the Laplace-type operator $L_\Pi = A_g$ that enters the present paper. This identification is now unconditional: Q9 [8] proves hypothesis [H-lift], and Q8 [18] together with Q10 [19] establish the metric coefficients $A_Z = A_H = 2$, giving $g^{\mu\nu} = \operatorname{diag}(-A_\tau, 2, 2, 2)$ with only the temporal coefficient A_τ remaining to be determined. Numerical computation of the admissibility weight $a_q(s) = \mathbb{E}_{n,c}\big[\|(\rho_q(s) - I)\Pi_q v_c^{(n)}\|^2_{C_q}\big]$ for all conjugate pairs $(c, q-c)$ at $q \in \{29, 61, 101, 151, 211\}$ (267 valid pairs in total) yields the cross-q summary

$$a_q(s)\cdot q^2 \;\in\; \{2.002,\ 1.988,\ 1.982,\ 1.972,\ 1.983\},$$

with inter-pair standard deviations below 0.09 and mean isotropy $|a_q(X) - a_q(Y)|\cdot q^2 < 0.07$ across all primes. A small number of non-generic pairs (those whose character c has special arithmetic properties modulo q) exhibit larger anisotropy and account for the residual variance; generic pairs satisfy $A_{\text{hw}} \approx 2.000$ to three significant figures. Hypothesis H-w of [6] is thereby confirmed numerically across five primes: $a_q(s)\cdot q^2 \to 2$ as $q \to \infty$, with A independent of s. The isotropy $a_q(X) = a_q(Y)$ implies that the Dirichlet form limit is isotropic, consistent with the emergence of a Lorentzian metric. The Einstein response derived here is therefore, at the structural level, a consequence of Heisenberg torsion rather than of a postulated spacetime curvature.

Within the spectral framework considered in this work, the geometric response arises from the renormalized variation of the functional

$$S_\Pi[g] = \frac{1}{2}\log\det\{\}' A_g.$$

The heat-kernel analysis shows that the infrared-dominant local contribution to the metric variation is proportional to the Einstein tensor. The spectral construction therefore provides an explicit realization of the effective dynamics compatible with a non-injective projective structure.

Taken together, these results suggest the following structural picture. Non-injective projection allows the emergence of non-trivial fiber curvature, which in turn enables an effective geometric response. The spectral entropy functional provides a minimal mechanism realizing this response and selects the Einstein sector as the dominant infrared contribution to the effective dynamics. This mechanism is consistent with the relational spectral framework of [4], in which spacetime geometry emerges operationally from relational configurations under non-injective projection. The same non-injectivity argument further yields gauge charges and gauge groups as fibre invariants of Π, establishing Yang–Mills structure without additional postulates [16].

7.3 Relation to induced gravity

The mechanism identified here is structurally identical to induced gravity in the sense of Sakharov. The gravitational coupling emerges from renormalization of the geometric sector. The spectral cutoff scale sets the magnitude of the induced Einstein–Hilbert term.

No assumption of fundamental gravitational dynamics is made. The geometric response arises from operator-theoretic structure.

7.4 Limitations

Several limitations must be emphasized.

First, the derivation is performed in a Euclidean Riemannian framework. Although the variational structure of the spectral functional is well defined in this setting, a complete treatment of causal propagation requires a Lorentzian formulation of the theory.

Second, non-local contributions generated by the spectral expansion are not eliminated. While they are suppressed in the infrared regime considered here, they may become relevant at higher curvature or shorter scales.

Third, the precise numerical value of the induced Newton constant depends on the ultraviolet completion and on the detailed matter content contributing to the spectral functional. The uniqueness result of Section 6 constrains the functional form of this completion but does not fix its overall normalisation independently of the matter content.

Fourth, Hypothesis [H-ext] (Section 6.1) has not yet been derived analytically from the Foundation axioms. The uniqueness of the Born–Infeld tensorial completion is conditional on this hypothesis.

Finally, the present work focuses on the emergence of the effective gravitational field equations. Questions of dynamical stability, cosmological evolution, and coupling to additional matter sectors lie beyond the scope of the present analysis.

7.5 Remark on geometric flows and spectral monotonicity

The functional S_Π depends on the full eigenvalue spectrum of the elliptic operator A_g. Under geometric evolution equations such as the Ricci flow, spectral quantities associated with the Laplacian exhibit monotonic behavior.

In particular, monotonicity of the first eigenvalue $\lambda_1(-\nabla_g^2)$ of the scalar Laplacian under Ricci flow has been established by Ma [20]. Related results for modified Laplace-type operators involving curvature terms were obtained by Cao [21]. These developments are rooted in the entropy framework introduced by Perelman [22].

Although S_Π is not identical to Perelman's entropy functional, its spectral definition suggests a structural relation between geometric flows and spectral entropy variation. Clarifying this relation would require a dedicated analysis and lies beyond the scope of the present work.

7.6 Outlook

The analysis presented here shows that a minimal spectral entropy functional generates, after renormalization, an infrared-dominant Einstein response.

Several directions follow naturally from this result.

- Extension of the spectral framework to Lorentzian signature, allowing a fully causal formulation of the effective dynamics and the study of propagating gravitational modes.
- Coupling of the spectral sector to additional operator structures describing matter degrees of freedom, leading to a unified description of geometric and material contributions to the effective stress tensor.
- Investigation of regimes in which higher-order curvature contributions become relevant. Section 6 identifies the unique admissible non-perturbative completion in the Born–Infeld class; the resulting strong-curvature dynamics and its predictions for gravitational collapse, bouncing cosmologies, and singularity resolution remain to be explored.
- Exploration of the relation between projection structure and emergent geometry, clarifying how non-injective effective descriptions may generate curvature responses in the infrared regime.
- Completion of the derivation chain $\Pi_q \to \mathcal{E}_q \to L_\Pi = A_g \to G_{\mu\nu}$: the identification $L_\Pi = A_g$ is now unconditional (Q9 [8]), and the metric coefficients are established as $g^{\mu\nu} = \mathrm{diag}(-A_\tau, 2, 2, 2)$ (Q8 [18], Q10 [19]). Determining the temporal coefficient A_τ from the projective temporal ordering data of [6] is the remaining open step.

More broadly, the present result supports the idea that geometric gravity may arise as a low-energy manifestation of spectral operator structure rather than as a fundamental dynamical input.

References

[1] Beau, J.: Cosmochrony: A Non-Injective Projection Theory of Emergent Physics. original white paper, doi:10.5281/zenodo.17957509 (2026)

[2] Vassilevich, D.V.: Heat kernel expansion: user's manual. Phys. Rept. **388**, 279–360 (2003) https://doi.org/10.1016/j.physrep.2003.09.002

[3] Parker, L., Toms, D.: Quantum Field Theory in Curved Spacetime. Cambridge University Press, ??? (2009)

[4] Beau, J.: Relational Reconstruction of Spacetime Geometry from Graph Laplacians. Cosmochrony companion paper A, doi:10.5281/zenodo.18356037 (2026)

[5] Beau, J.: Non-Injectivity as a Structural Necessity of Genuine Emergence: A No-Go Theorem for Faithful Emergent Descriptions. Cosmochrony companion paper "ENI" or L, doi:10.5281/zenodo.19391746 (2026)

[6] Beau, J.: Large-q Limit of the Admissible Fibre: Representational and Operator Convergence. spectral admissibility subprogram "Q5a" paper, doi:10.5281/zenodo.19642369 (2026)

[7] Beau, J.: BFS Shell Stratification and the Emergence of Four-Dimensional Lorentzian Geometry. spectral admissibility subprogram "Q5b" paper, doi:10.5281/zenodo.19686700 (2026)

[8] Beau, J.: Proof of the Lifting Hypothesis [H-lift]: Kinetic-Sector Identification via Generator Suppression. "Q9" paper, doi:10.5281/zenodo.19880574 (2026)

[9] Seeley, R.T.: Complex powers of an elliptic operator. Proc. Symp. Pure Math. **10**, 288–307 (1967)

[10] DeWitt, B.S.: Dynamical theory of groups and fields. Gordon and Breach (1965)

[11] Sakharov, A.D.: Vacuum quantum fluctuations in curved space and the theory of gravitation. Sov. Phys. Dokl. **12**, 1040 (1968)

[12] Barvinsky, A.O., Vilkovisky, G.A.: The generalized schwinger–dewitt technique. Phys. Rept. **119**, 1–74 (1985)

[13] Barvinsky, A.O., Vilkovisky, G.A.: Covariant perturbation theory. Nucl. Phys. B **333**, 471–511 (1990)

[14] Birrell, N.D., Davies, P.C.W.: Quantum Fields in Curved Space. Cambridge University Press, ??? (1982)

[15] Beau, J.: Bounded Relaxation and the Dynamical Selection of Spacetime Geometry. Cosmochrony companion paper C about Born-Infeld, doi:10.5281/zenodo.18407505

(2026)

[16] Beau, J.: Admissible Non-Injective Transitions as the Primitive of Physical Description. Cosmochrony Foundation Paper "M", doi:10.5281/zenodo.19616329 (2026)

[17] Beau, J.: Colour Triplet Co-admissibility on $\mathrm{Heis}_3(\mathbb{Z}/q\mathbb{Z})$: Numerical Test of Hypothesis [H-color]. "O32" paper, doi:10.5281/zenodo.20145607 (2026)

[18] Beau, J.: Sub-Principal Symbol of the Effective Operator and Casimir Rigidity of the Central Direction. "Q8" paper, doi:10.5281/zenodo.19879909 (2026)

[19] Beau, J.: Asymptotic $\mathfrak{su}(2)$-Isotropy of the Effective Quadratic Form and the Identification $A_H = 2$. "Q10" paper, doi:10.5281/zenodo.19880900 (2026)

[20] Ma, L.: Eigenvalue monotonicity for the ricci flow. Ann. Global Anal. Geom. **29**, 287–293 (2006)

[21] Cao, X.-D.: Eigenvalues of $(-\delta + \frac{R}{2})$ on manifolds with nonnegative curvature operator. Math. Ann. **337**, 435–441 (2007)

[22] Perelman, G.: The entropy formula for the ricci flow and its geometric applications. arXiv:math/0211159 (2002)